

Ninety-Three Per Cent of a Backlog Is Not a Backlog: Reading a Public Publisher's Own Quality Signal

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Abstract

Compliance systems, regulatory technology and models grounded in statutory text now read published law directly. A companion study proposed a five-field provenance schema so that a consumer of such data can tell what stands behind each record; this article applies it prospectively to a second jurisdiction and reports what happened when it did. The United Kingdom's official legislation service publishes, per Act and machine-readably, the amendments it has not applied to the text it displays: 14,407 across the Public General Acts of 1988–2026. Read as a backlog — the reading the flag invites, and the one this study first recorded — that overstates by a factor of 131. 93.0 per cent are prospective, enacted but not commenced; a further 901 carry a commencement date still in the future, and the service is right not to apply either. What remains is 99 effects in 9 Acts, none in force longer than 21 days against a published three-month target, all from a single commencement three weeks before the analysis date. The service meets its own standard without exception. The transferable result is the error: everything needed to read the record correctly is published, but the flag inviting the wrong conclusion is an attribute on each effect while the fact correcting it is a child element inside it, and the publisher's prose covers all four states with one term. A consumer who parses attributes and checks the documentation is confirmed in a false accusation against a service doing its job. Publishing the distinguishing fact is not enough; its placement decides what most readers conclude.

Keywords: public data infrastructure; data quality; open government data; metadata; legal informatics; provenance

1 Introduction

A companion study measured whether one state's official legal record was available and internally consistent when someone tried to build on it, and found that neither could be assumed.¹ It proposed a five-field provenance schema for publishers of official data, and stated a weakness plainly: one jurisdiction, and an argument rather than a demonstration that the method travels.

This article is the demonstration, and what it demonstrates is not what it set out to. Repeating a measurement elsewhere tests whether a number recurs. Applying an instrument elsewhere tests the instrument. The instrument failed first, in a way worth reporting.

The United Kingdom's official legislation service publishes, in each Act's own metadata, a machine-readable list of amendments it has not applied to the text it displays. Read as a backlog, that list says 14,407 amendments across the modern statute book are missing from the law as published — an unusually clean finding, and the one this study first recorded.

It is wrong by a factor of 131, and it took two corrections rather than one to find out — which is itself part of the argument. Each effect also carries whether the amendment is in force. 13,396 of them (93.0 per cent) are prospective: enacted, not commenced, no date set. A further 901 have a commencement date

¹Cited as a companion study so this article can be read on its own; the corpus and analysis are openly published.

still in the future. The service is right not to apply either — applying an amendment that is not in force would misstate the law. What remains is 99 effects, and none has been in force for more than 21 days against a published target of three months.

The second of those categories is why one correction was not enough. Excluding the prospective effects is the obvious fix once the in-force field is found, and it still overstates by an order of magnitude: an effect can be non-prospective — its commencement settled — with that commencement fixed for a date that has not arrived. A date set for next spring is no more in force than no date at all. Each stage of this measurement looked correct until the next field was read.

The finding, then, is the opposite of the one this article was drafted to report: the service meets its own standard without exception. But the more useful result is the error. The distinguishing field is a child element inside each effect; the flag that invites the wrong reading is an attribute on the effect itself, and the publisher’s own prose calls both ‘unapplied effects’ without separating them. A reader who takes the obvious path — parse the attributes, trust the wording — produces a false accusation against a public service that is doing its job, and produces it with every appearance of rigour.

That is the companion study’s argument arriving from an unexpected direction. That paper asked publishers to ship the strength of the evidence with the text. Here the evidence is shipped, completely and machine-readably, and the default reading of it is still wrong by two orders of magnitude. Publishing the distinguishing fact is not sufficient; where it sits, relative to the fact that invites the error, decides what most readers will conclude.

2 What the service publishes about its own currency

Two features of `legislation.gov.uk` matter here, and both are the publisher’s own statements rather than this study’s inferences.

The first is the *unapplied effect*. When an amendment has been made to an Act and the revised text has not yet been changed to reflect it, the service records the fact in the Act’s metadata and flags it to readers: ‘If the primary legislation on `legislation.gov.uk` has any unapplied effects, we flag them in the “Changes to Legislation” banner on the website. Clicking on the banner reveals the outstanding changes.’² Each such effect names the amending instrument, the provision affected, and the kind of change.

The second is a service standard: ‘We aim to incorporate new amendments into the text of the legislation within three months of those amendments coming into force.’³

The standard is what makes compliance measurable rather than a matter of impression, and it also rules something out. A three-month target means the expected steady state is not zero unapplied effects: a queue is the design, not the defect, so the existence of flagged effects establishes nothing on its own. What the standard permits is a direct test — how long has each amendment actually been in force without being applied — and the answer needs the commencement date, which the service publishes for every effect.

3 Method

Collection. Every Public General Act from 1988 to 2026 was enumerated from the service’s own year feeds, and each Act’s metadata retrieved from its `introduction/data.xml` endpoint. That endpoint carries the complete unapplied-effects block — verified rather than assumed, by comparing it against the full text and the contents listing for the same Act and finding the same effects in all three — at a fraction of the bytes. The service’s `robots.txt` declares a five-second crawl delay and the collector honours it, which is why a sweep of the statute book takes hours.

²*Understanding legislation*, <https://www.legislation.gov.uk/understanding-legislation>, under the heading ‘Our editorial practice and timescales’. The heading matters: the same page carries a separate section on legislation originating from the EU, and the commitment quoted here sits outside it and is stated generally.

³*Ibid.*, same heading.

The schema, populated at collection time. The companion study’s five fields were written for each record as it was retrieved, unchanged from that paper’s definition. This is the point of the exercise rather than decoration: that paper claims the schema costs nothing because a collector that knows how it obtained a document already knows all five values. Populating them prospectively is the only way to find out whether that is true.

What is measured, and what is not. The companion study measured two properties: availability and internal consistency. Only the second is compared here, and the reason is structural rather than practical. Four of the schema’s five fields describe the *encounter* — whether the live source answered this collector, on this network, on this day — and encounters do not compare across collections, because they differ before the records do. No control corrects for that. Availability figures are recorded and published for this collection, and are not set against the companion study’s.

An unapplied effect has no such problem *in principle*: it is a property of the data as published, so any reader, on any network, at any time, can obtain the same count. In practice they will not, and the rest of this article is about why — the count depends on which of the record’s fields the reader consults, and the obvious selection is wrong by two orders of magnitude. Reproducibility of a measure and agreement between measurers are different properties, and only the first is guaranteed here.

4 Results

Coverage. 1,578 Public General Acts from 1988 to 2026 were listed and 1,578 retrieved; none was lost.

What the flag contains. Of the 14,407 effects the service flags as not applied:

	Effects	Share
Prospective — enacted, no commencement date	13,396	93.0%
Commencement scheduled, date still future	901	6.3%
In force now, not applied	99	0.7%
In force, no date recorded	11	—

Table 1: The same flag, four meanings. Only the last two describe text that does not reflect the law.

The measure that matters. 9 Acts of 1,578 — 0.6 per cent — display text that does not reflect the law in force, across 77 distinct provisions. Against a body of 101,711 paragraphs, that is 0.8 affected provisions per thousand.

Compliance is total, and concentrated in one event. The service undertakes to incorporate amendments within three months of their coming into force. 0 effects exceed that. The longest any amendment in the modern statute book has been in force without being applied is 21 days.

The dated in-force effects are not spread across the period at all. All 99 came into force on the same day, three weeks before the analysis date, and all belong to two related instruments: the Pension Schemes Act 2021 and the Pension Schemes (Northern Ireland) Act 1993.⁴ The entire population of out-of-date text in the modern UK statute book is one commencement, three weeks old.

That is why this article carries one figure and not two. A distribution of 99 identical values is not a chart.

⁴The remaining 11 effects are recorded as in force without a date and cannot be timed; they are counted in the affected totals and excluded from the duration figures.

What a flag of “not applied” actually means UK Public General Acts, 1988–2026

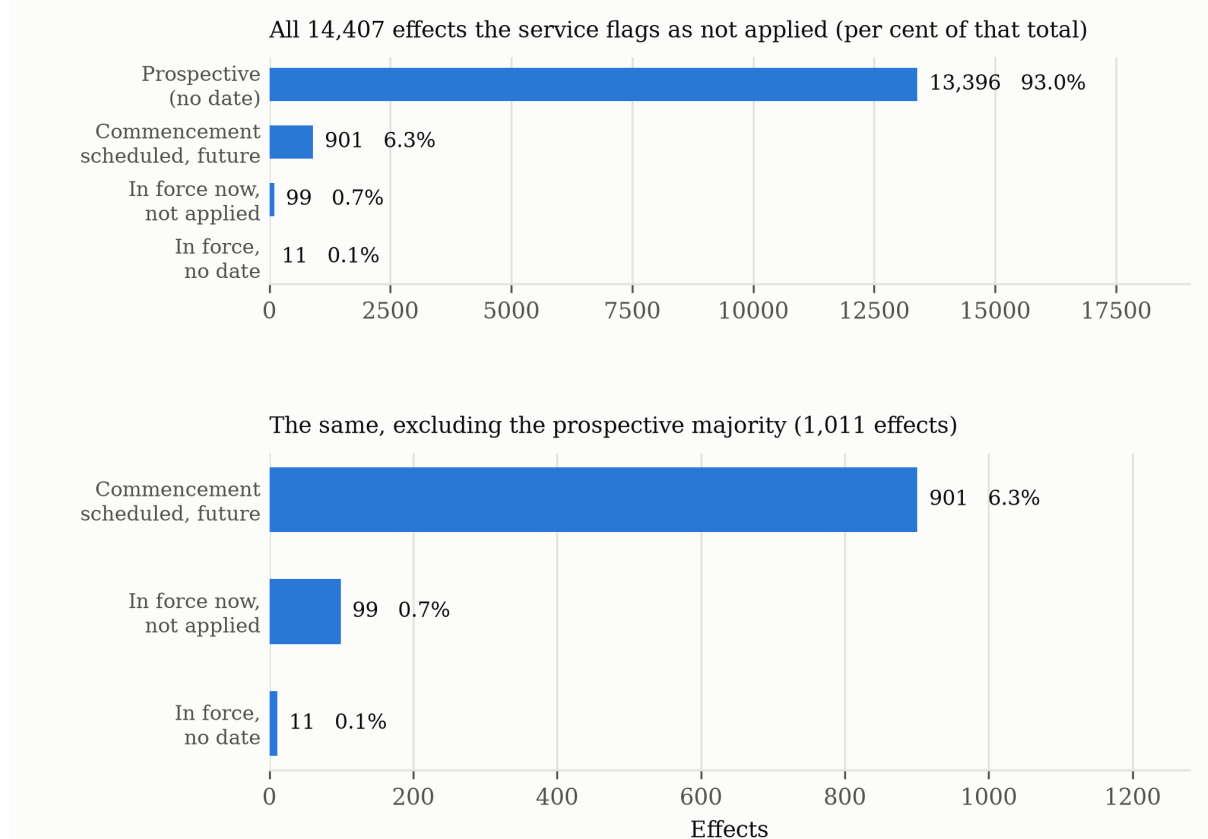


Figure 1: The composition on a linear scale, then the same excluding the prospective majority so that the smaller categories are legible. A logarithmic axis would have fitted all four comfortably and removed the disproportion, which is the finding.

5 What the schema learned about itself

Two changes to the companion study’s proposal came out of using it, and neither would have come out of thinking about it.

A discrepancy has to record who declared it. Applied to a well-run interface, four of the five fields take a single value across every record in the collection: one source class, one retrieval route, one corroboration count, one transformation. That is the schema working, not failing — it costs nothing here because there is nothing to say.

But the defect this study measures was invisible to all five fields. Every affected Act was retrieved perfectly — official primary source, live, no transformation, nothing to report — so under the schema as written an Act displaying text that does not reflect the law is indistinguishable from one that is entirely current. The argument does not depend on the size of the defect, and the size is small: 9 Acts, 110 effects. An earlier draft made the same point with an Act omitting hundreds of amendments, which the corrected measure shows does not exist.

The remedy is not a sixth field. The purpose the companion study gives discrepancy is ‘is there a known problem with this record?’, and a publisher flagging its own text as not current is exactly that. What the field’s *values* omit is that a discrepancy can be declared by the source as well as found by the collector — and a null means different things in the two cases. Found by a collector, it cost two or more sources and a comparison, so a null may only mean nobody looked hard enough. Declared by a source, it

costs nothing to record, so a null means the publisher did not say. A single null standing for both is the collapse that paper objects to when one confidence score stands for availability and consistency at once.

The two jurisdictions are the argument. Here, 9 records carry a source-declared discrepancy and 0 a collector-found one. In the companion corpus it was the reverse: four self-contradicting instruments, every one of them found by hand. The same class of defect, with opposite provenance — and a straightforward consequence for who should be doing the work.

The fields divide by what they describe, not by what they are about. `source_class` describes the record: whose copy this is. `retrieval_route`, `corroboration` and `transformation` describe the encounter: whether the live source answered *us*, how many sources *we* found agreeing, what *we* had to do to the bytes we got. `discrepancy` is either, depending on who declared it.

Four of five describe the encounter, which is why the availability half of this comparison could not be salvaged. It also complicates the companion study’s own recommendation. That paper asks publishers to populate these fields; if a publisher does, the encounter fields describe the publisher’s encounter with its own source, which is a third thing again and comparable with neither. A provenance schema intended for use across collections has to mark which of its fields are encounter-relative. That one does not.

6 What follows

Publishing the distinguishing field is not enough; placing it is. Everything needed to read this record correctly is published. The failure is one of arrangement: the flag that invites the wrong conclusion is an attribute on the effect, and the fact that corrects it is a child element inside it. A consumer parsing attributes — the ordinary way to read this format — gets the first and not the second, and nothing in the response marks the omission. A publisher cannot control how it is read, but it can decide which fact is cheapest to reach.

Prose that does not distinguish what the data distinguishes. The service describes flagging ‘unapplied effects’. Its data separates four states under that phrase, only two of which mean the text is out of date. The documentation was checked against the reading used here and appeared to confirm it, because the words are true of all four. Where a vocabulary is coarser than the schema it describes, the documentation will validate a wrong reading.

A measurement that reverses is worth more than one that confirms. This study set out expecting to find a backlog, found one, and had to withdraw it. The corrected result — a public publisher meeting its stated standard without exception across 1988–2026 — is a kind of finding the open-data literature reports rarely, in part because it is not what anyone goes looking for.

7 Limits

A claim this article previously made, withdrawn. An earlier draft stated that commencement dates are not published and concluded that the publisher withholds the field needed to verify its own service standard. That was the opposite of true. Every effect carries its commencement position in `ukm:InForce`, and the durations reported above are a subtraction over that field. The error had the same cause as the headline error: a parse that read attributes and not child elements, and a conclusion drawn from what the parse returned rather than from what the response contained.

Denominators are choices, and all of them are reported. An Act the service serves as enacted rather than in revised form has no revised text for an amendment to be missing from; a wholly repealed Act has no live text at all. Excluding them raises the affected share and including them lowers it. Neither is obviously right, so the article reports the share on all three bases rather than choosing one.

Coverage. 1,578 Acts were listed and 1,578 retrieved. Retrieval failures were recorded rather than retried away while availability was in scope; once it left the comparison, a second pass recovered what it could, and both attempts are kept in the record.

The modern statute book, and a lower bound. The sweep covers Public General Acts from 1988 to 2026. That range was chosen on the assumption that unapplied effects accumulate in modern legislation, and the assumption is wrong: a sample of twenty Acts drawn from before it found five carrying unapplied effects and 218 effects in total, the Highways Act 1980 alone accounting for 132. Every total reported here is therefore a lower bound on the statute book as a whole rather than a census of it, and the concentration figures are computed within the period covered.

Extending the sweep is a matter of time rather than method — at the publisher’s declared crawl delay the pre-1988 book is many hours of collection — and the range is stated rather than quietly presented as complete.

Primary legislation only. Statutory instruments are not covered. They are the larger population and carry their own unapplied effects; nothing here should be read as a statement about them.

Declarations

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Competing interests. The author declares none.

Ethics approval. Not applicable. The study involves no human participants, no animal subjects and no personal data; it analyses published national legislation and its publisher’s own metadata.

Declaration of generative AI use. *Tool:* Claude Opus 5 (Anthropic). *How it was used:* to write the collection, analysis, verification and figure scripts that produce every number reported here, to draft and revise the manuscript, and to run adversarial checks against its own claims. *Reason for use:* the study required retrieving and parsing the metadata of the entire modern statute book under a five-second crawl delay and auditing the result, which is not feasible by hand at that scale for a single researcher. The author set the research question, directed the analysis, verified the outputs against the underlying records, and is responsible for the content.

CRedit roles. A. Almohammed: conceptualisation, data curation, methodology, software, formal analysis, investigation, writing — original draft, writing — review and editing.

Data availability. The collection, the analysis and the scripts that reproduce every number and figure are openly available at <https://github.com/Abdullah-m7/saudi-legal-corpus-ai>. The underlying metadata belongs to `legislation.gov.uk` and is published by the Crown under the Open Government Licence.

A note on the subject. This article measures a defect that `legislation.gov.uk` discloses about itself. It is not an accusation. Almost no comparable publisher discloses as much, which is what makes the measurement possible at all, and the article’s argument is that disclosure of this quality still leaves a reader unable to answer the question it raises.